

Parallel-plate capacitor: field, charge and capacitance (clean 1-D argument)

Set-up (ideal capacitor). Two large conducting plates of area A separated by a small gap d (air/vacuum). Top plate is held at potential $\phi = V$, bottom plate is grounded ($\phi = 0$). If plate dimensions $\gg d$, the fringing field is negligible and the field is approximately uniform in $0 < z < d$.

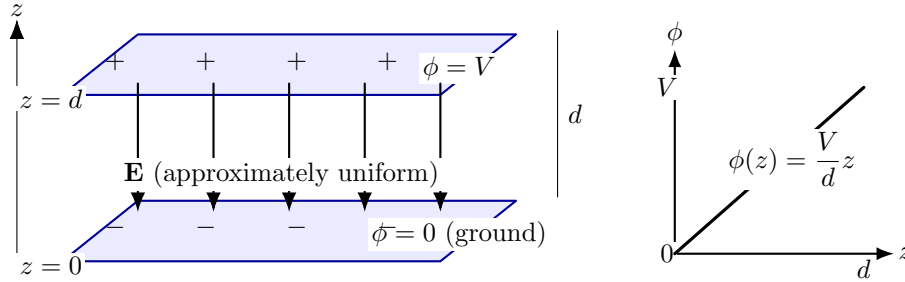


Figure 1: Ideal parallel-plate capacitor: if plate size $\gg d$, the field in $0 < z < d$ is approximately uniform and fringing is negligible.

Step 1: Why a 1-D (electrostatic) model is valid in the gap

- The plates are conductors $\Rightarrow$ each plate is an equipotential surface.
- In the central region (away from edges), the geometry is translationally symmetric in x, y and the field lines are approximately straight *from one plate to the other*.
- Inside the dielectric (air/vacuum) between plates, there is no volume charge: $\rho = 0$ for $0 < z < d$.

Step 2: Use Poisson/Laplace equation to get the potential

Poisson's equation is

$$\nabla^2 \phi = -\frac{\rho}{\epsilon_0}.$$

In the region between plates $\rho = 0$, so

$$\nabla^2 \phi = 0.$$

With the 1-D approximation ($\phi = \phi(z)$),

$$\frac{d^2 \phi}{dz^2} = 0, \quad 0 < z < d,$$

and the boundary conditions are

$$\phi(0) = 0, \quad \phi(d) = V.$$

Integrating,

$$\phi(z) = \frac{V}{d} z$$



Step 3: Electric field

$$\mathbf{E} = -\nabla\phi = -\frac{d\phi}{dz}\hat{\mathbf{z}} = -\frac{V}{d}\hat{\mathbf{z}} \Rightarrow \boxed{\mathbf{E} = -\frac{V}{d}\hat{\mathbf{z}} \text{ (constant in the gap).}}$$

Step 4: What is the charge on the plate? (Gauss pillbox)

Use a small *Gaussian pillbox* straddling the conductor surface. Inside a conductor, $\mathbf{E} = \mathbf{0}$, so the inner face contributes zero flux. On the curved surface, $\mathbf{E}$ is tangential, so $\mathbf{E} \cdot d\mathbf{S} = 0$ there. Thus only the outer flat face contributes.

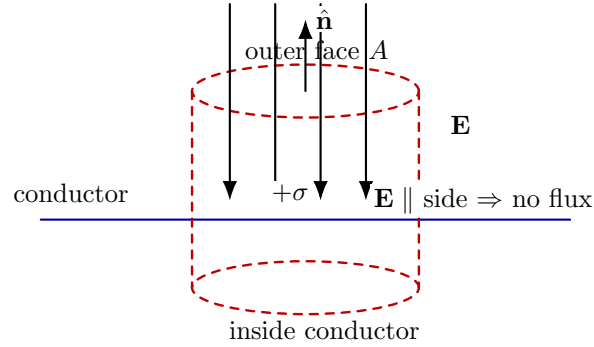


Figure 2: Gaussian pillbox at the conductor surface: only the *outer face* contributes to flux (curved side: tangential field; inner face: inside conductor $\mathbf{E} = \mathbf{0}$).

Apply Gauss's law:

$$\oiint \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enc}}}{\epsilon_0}.$$

Only the outer face contributes, and $\mathbf{E}$ is normal to that face, so

$$EA = \frac{\sigma A}{\epsilon_0} \Rightarrow \boxed{\sigma = \epsilon_0 E}$$

and with $E = \frac{V}{d}$ in magnitude,

$$\boxed{\sigma = \epsilon_0 \frac{V}{d}, \quad Q = \sigma A = \epsilon_0 \frac{A}{d} V.}$$

Step 5: Relate charge and voltage $\Rightarrow$ capacitance

In the same central-region approximation, the surface charge densities are $\pm\sigma$ and the field in the gap obeys

$$E = \frac{\sigma}{\epsilon_0} \Rightarrow \sigma = \epsilon_0 \frac{V}{d}.$$

Total charge on the plate is $Q = \sigma A$, hence

$$Q = \varepsilon_0 \frac{A}{d} V \quad \Rightarrow \quad \boxed{C \equiv \frac{Q}{V} = \varepsilon_0 \frac{A}{d}}$$

(For a dielectric with permittivity ε , replace $\varepsilon_0 \rightarrow \varepsilon$.)

Step 6: Energy stored in a capacitor (work-done picture)

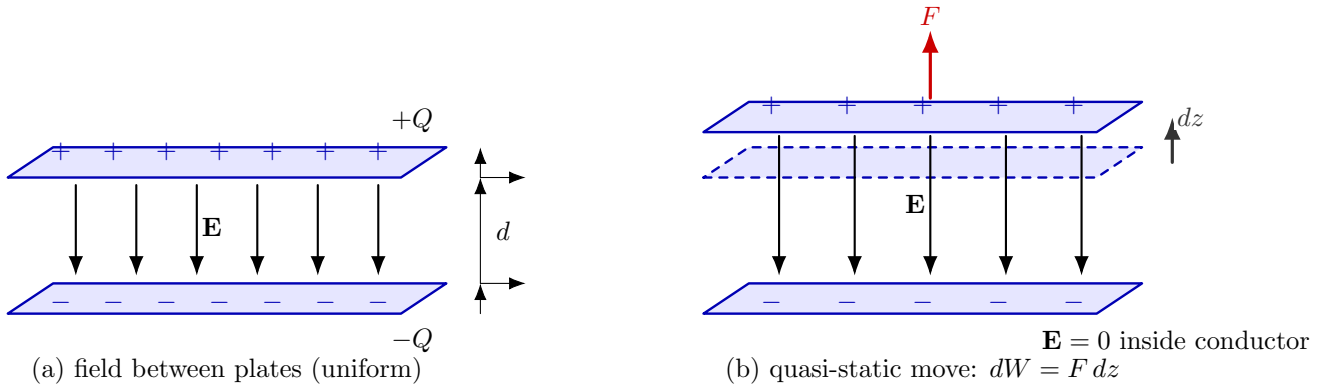


Figure 3: Energy stored in a capacitor via work done: the force on the plate uses the average field at the conductor surface, giving $F = \frac{1}{2}EQ = \frac{1}{2}\varepsilon_0 E^2 A$, and $W = \int_0^d F dz = \frac{1}{2}QV = \frac{1}{2}CV^2$.

Consider two large conducting plates of area A separated by distance d (vacuum/air), with charges $+Q$ and $-Q$ (fringing neglected, i.e., $A \gg d^2$). The field between the plates is approximately uniform:

$$\sigma = \frac{Q}{A}, \quad E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}, \quad V = \int_0^d E dz = Ed, \quad C = \frac{Q}{V} = \frac{\varepsilon_0 A}{d}.$$

Key idea for force on a conducting surface. Inside a conductor, $\mathbf{E}_{\text{in}} = \mathbf{0}$. Just outside the surface (in the gap), $E_{\text{out}} = \sigma/\varepsilon_0$ (normal to the surface). The force on surface charges uses the *average* field at the surface:

$$\mathbf{E}_{\text{avg}} = \frac{\mathbf{E}_{\text{out}} + \mathbf{E}_{\text{in}}}{2} = \frac{1}{2}\mathbf{E}_{\text{out}}.$$

Hence, the magnitude of force on the plate is

$$F = Q E_{\text{avg}} = Q \left(\frac{E}{2} \right) = \frac{1}{2}EQ.$$

Work done $\Rightarrow$ stored energy. Now move the upper plate *quasi-statically* from $z = 0$ to $z = d$ while keeping Q fixed. Since $E = Q/(\varepsilon_0 A)$ is constant (independent of d for fixed Q),

$$dW = F dz = \left(\frac{1}{2}EQ \right) dz.$$



Therefore the energy stored (work required to build the configuration) is

$$W = \int_0^d \frac{1}{2}EQ \, dz = \frac{1}{2}EQd = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{Q^2}{2C}.$$

Energy density. The energy is stored in the electric field in the volume between plates (Ad):

$$u \equiv \frac{W}{Ad} = \frac{\frac{1}{2}EQd}{Ad} = \frac{\frac{1}{2}E(\epsilon_0 AE)d}{Ad} = \frac{\frac{1}{2}\epsilon_0 E^2 Ad}{Ad} = \boxed{\frac{1}{2}\epsilon_0 E^2}.$$

More generally, for any electrostatic field distribution,

$$U = \frac{1}{2} \iiint_{\text{all space}} \epsilon_0 |\mathbf{E}|^2 \, dV.$$

Key takeaways (parallel-plate capacitor)

- Conducting plates are equipotential: $\phi(0) = 0$, $\phi(d) = V$.
- In the central region (fringing neglected): $\phi(z) = \frac{V}{d}z$ and $E = \frac{V}{d}$.
- Charge–voltage relation: $Q = \epsilon_0 \frac{A}{d}V \Rightarrow C = \epsilon_0 \frac{A}{d}$.
- Stored energy: $U = \frac{1}{2}CV^2$ (equivalently $u = \frac{1}{2}\epsilon_0 E^2$ in the gap).

Capacitance of a Co-axial Cable

Geometry and boundary conditions. Two long, concentric *conducting* cylinders:

- Inner conductor radius a is held at potential $\phi(a) = V_0$.
- Outer conductor inner radius b is grounded: $\phi(b) = 0$.
- Region $a < r < b$ is air/vacuum (no free charge in the dielectric).
- Cylinders are long $\Rightarrow$ no variation along z , and cylindrical symmetry $\Rightarrow$ no variation with θ .

Hence $\phi = \phi(r)$ only, in the dielectric region $a < r < b$.

Step 1: Poisson/Laplace equation in the dielectric

Start from Poisson's equation:

$$\nabla^2 \phi(\mathbf{r}) = -\frac{\rho(\mathbf{r})}{\epsilon_0}.$$



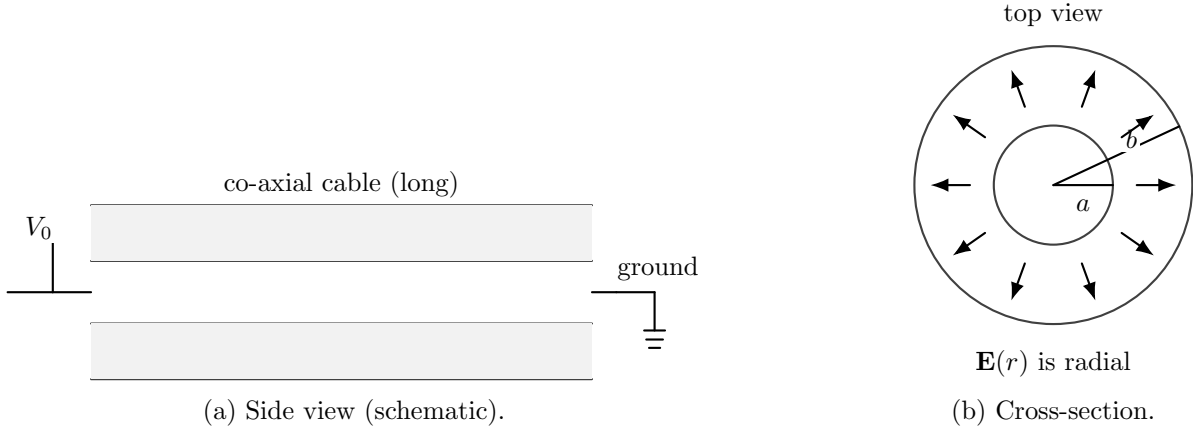


Figure 4: Co-axial cable: inner conductor at $\phi = V_0$, outer conductor grounded. Field exists only in the dielectric region $a < r < b$.

In the region $a < r < b$ (air/vacuum), there is *no free charge density*:

$$\rho = 0 \quad \Rightarrow \quad \nabla^2 \phi = 0 \quad (a < r < b).$$

By symmetry for a long co-axial cable: $\phi = \phi(r)$ only. In cylindrical coordinates, with only r -dependence:

$$\nabla^2 \phi = \frac{1}{r} \frac{d}{dr} \left(r \frac{d\phi}{dr} \right) = 0, \quad a < r < b.$$

Step 2: Solve for $\phi(r)$

Integrate once:

$$\frac{d}{dr} \left(r \frac{d\phi}{dr} \right) = 0 \quad \Rightarrow \quad r \frac{d\phi}{dr} = k \quad \Rightarrow \quad \frac{d\phi}{dr} = \frac{k}{r}.$$

Integrate again:

$$\phi(r) = k \ln r + A.$$

Apply boundary conditions (outer grounded, inner at V_0):

$$\phi(b) = 0 \Rightarrow 0 = k \ln b + A \Rightarrow A = -k \ln b,$$

$$\phi(a) = V_0 \Rightarrow V_0 = k \ln a - k \ln b = k \ln \left(\frac{a}{b} \right).$$

Hence

$$k = \frac{V_0}{\ln(a/b)}.$$

So the potential in $a < r < b$ is

$$\boxed{\phi(r) = \frac{V_0 \ln(r/b)}{\ln(a/b)}, \quad a < r < b.}$$

Step 3: Electric field $\mathbf{E}(r)$

$$\mathbf{E}(r) = -\nabla\phi = -\frac{d\phi}{dr} \hat{\mathbf{r}}.$$

Differentiate:

$$\frac{d\phi}{dr} = \frac{V_0}{\ln(a/b)} \cdot \frac{1}{r}.$$

Therefore,

$$\mathbf{E}(r) = -\frac{V_0}{r \ln(a/b)} \hat{\mathbf{r}} = \frac{V_0}{r \ln(b/a)} \hat{\mathbf{r}}, \quad a < r < b.$$

(Direction: for $V_0 > 0$ on the inner conductor and outer grounded, $\mathbf{E}$ points radially outward.)

Step 4: Charge per unit length and capacitance per unit length

Use Gauss's law with a coaxial Gaussian cylinder of radius r and length L (with $a < r < b$):

$$\oiint \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enc}}}{\epsilon_0} \Rightarrow E(r) (2\pi r L) = \frac{\lambda L}{\epsilon_0}.$$

So

$$E(r) = \frac{\lambda}{2\pi\epsilon_0 r}.$$

Compare with the derived field $E(r) = \frac{V_0}{r \ln(b/a)}$ to get

$$\lambda = 2\pi\epsilon_0 \frac{V_0}{\ln(b/a)}.$$

Hence the capacitance per unit length ($C' = \lambda/V_0$) is

$$C' = \frac{\lambda}{V_0} = \frac{2\pi\epsilon_0}{\ln(b/a)} \text{ (F/m)}.$$